DO NOW — RECALL

Lesson 090 · Inside City Hall: ALU, Control Unit and Regi

Name: _____ Class: _____ Date: _____

1. What does CU stand for?

2. What is the ALU?

3. What does ACC typically store?

4. A full adder has A=1,B=1 and carry-in=0. What are Sum and carry-out?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Control Unit.

2. Model answer

Arithmetic Logic Unit.

3. Model answer

Intermediate arithmetic/logic result.

4. Model answer

$1+1+0=10_2$, so Sum0 and carry-out1.

By the end, you can…

• Control Unit generates control signals and interprets the fetched instruction. It is not the component that performs arithmetic itself.

• ALU completes arithmetic and logic such as addition and comparisons; the ACC may hold a result in an accumulator-style model.

• Registers serve distinct roles: PC next instruction, MAR memory address, MDR transferred contents, CIR current instruction, ACC intermediate calculation.

Vocabulary — know these terms

Control Unit — Directs instruction execution.

ALU — Performs arithmetic and logic.

ACC — Intermediate calculation register.

PC — Address of next instruction.

NEW LEARNING

Read, annotate and explain

Observe

Control Unit generates control signals and interprets the fetched instruction. It is not the component that performs arithmetic itself.

Explain

ALU completes arithmetic and logic such as addition and comparisons; the ACC may hold a result in an accumulator-style model.

Evaluate

Registers serve distinct roles: PC next instruction, MAR memory address, MDR transferred contents, CIR current instruction, ACC intermediate calculation.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

ACC starts5; toy instruction ADD 7 is executed. Trace ALU and ACC.

Model response

ACC holds intermediate result, not memory address.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

For ACC=3 and toy ADD9, state ALU calculation, ACC result, CU role. [6]

Question 2

Explain different roles of PC, MAR and CIR in a memory fetch. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Pupil claims CU performs calculations while ALU stores next instruction address. Correct both errors. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

The Program Counter is the ALU register storing calculation results.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.